27 **C** $\frac{I_s}{I_p} = \frac{N_p}{N_s}$ and $\frac{N_p}{N_s} = \frac{V_p}{V_s}$

$$\frac{50}{2} = \frac{N_p}{50} \rightarrow N_p = 1250 \text{ turns}$$

$$\frac{N_p}{N_s} = \frac{V_p}{V_s} \rightarrow V_s = V_p \frac{N_s}{N_p} = 240\left(\frac{50}{1250}\right) = 9.6 \text{ V}$$

28 **D** Let p be the momentum of the electron